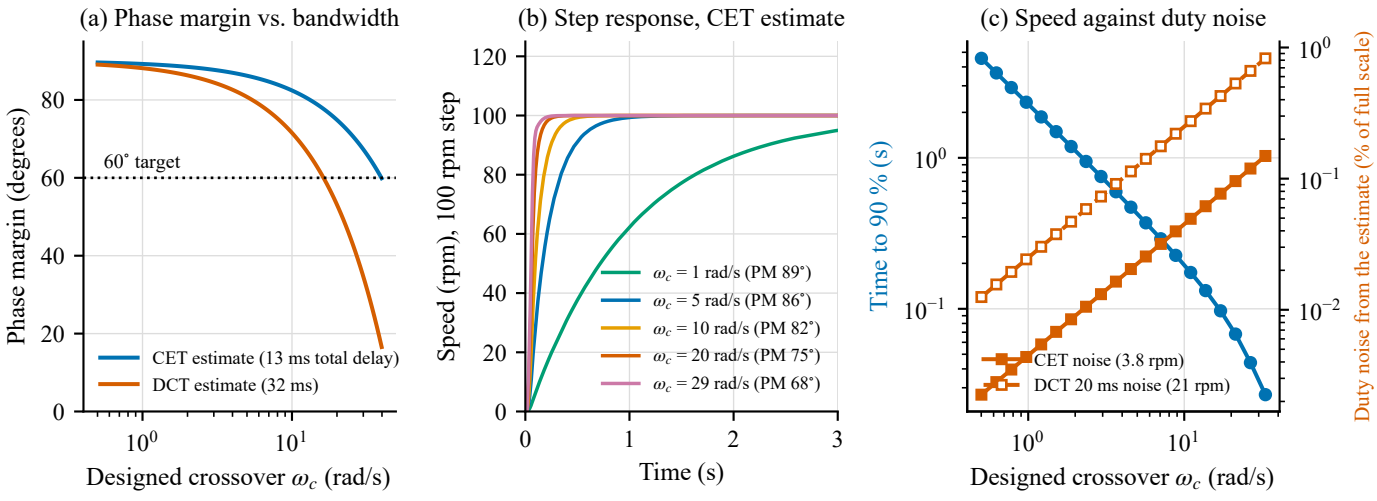

(a) Phase margin vs. bandwidth
Phase margin (degrees)
60° target
CET estimate (13 ms total delay)
DCT estimate (32 ms)
Designed crossover ω_c (rad/s)
(b) Step response, CET estimate
Speed (rpm), 100 rpm step
ω_c = 1 rad/s (PM 89°)
ω_c = 5 rad/s (PM 86°)
ω_c = 10 rad/s (PM 82°)
ω_c = 20 rad/s (PM 75°)
ω_c = 29 rad/s (PM 68°)
Time (s)
(c) Speed against duty noise
Time to 90 % (s)
Duty noise from the estimate (% of full scale)
CET noise (3.8 rpm)
DCT 20 ms noise (21 rpm)
Designed crossover ω_c (rad/s)